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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Sixth Semester B.Tech (Minor in Data Science) Degree Examinations

May 2025

IS1104-1 – WEB DATA MINING

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What is the main objective of association rule mining?**
A) Finding hidden relationships between items in a dataset B) Sorting data in a structured manner
C) Reducing data size for storage D) Predicting numerical values
2. **Which algorithm is most commonly used for mining association rules?**
A) Apriori B) K-Means
C) Decision Trees D) Naïve Bayes
3. **What happens if the minimum support threshold is set too high in association rule mining?**
A) Many potentially useful rules are ignored B) The dataset size increases
C) The number of frequent itemsets increases D) Computational time is reduced
4. **Which step follows frequent itemset generation in association rule mining?**
A) Rule extraction B) Data normalization
C) Clustering of similar rules D) Data transformation
5. **How is sequential pattern mining applied in web usage mining?**
A) By identifying common sequences of web pages visited by users B) By clustering users based on browsing behavior
C) By classifying web pages based on topic D) By ranking search engine results
6. **Which of the following algorithms is commonly used for sequential pattern mining in web usage analysis?**
A) GSP (Generalized Sequential Pattern) B) PageRank
C) Decision Trees D) K-Means
7. **How does sequential pattern mining help in website navigation optimization?**
A) By identifying common browsing paths and optimizing menu structures B) By reducing image sizes for faster page loading
C) By increasing the number of backlinks D) By ranking websites based on keyword density
8. **In web mining, what does a high-support sequential pattern indicate?**
A) A webpage with a high number of images B) A page with high ranking in search results
C) frequently shared web page on social media D) A common user navigation behavior across many sessions
9. **What is the purpose of indexing in information retrieval?**
A) To reduce the size of the document collection B) To store data in a compressed format
C) To improve the efficiency of searching by organizing information D) To generate summaries of documents

10. Which indexing technique is commonly used in search engines?
 A) Hash tables B) Inverted index
 C) Linked lists D) Decision trees
11. Which algorithm is used by Google to rank web pages based on link analysis?
 A) Apriori Algorithm B) PageRank Algorithm
 C) K-Means Clustering D) ID3 Decision Tree
12. Which measure combines precision and recall into a single metric?
 A) Cosine similarity B) PageRank score
 C) F1-score D) TF-IDF
13. What is Social Link Analysis used for?
 A) Analyzing social media posts B) Identifying relationships and patterns in
 C) Ranking web pages D) Detecting spam emails
14. What does a "hub" refer to in social link analysis?
 A) A node with the highest authority score B) A node that connects different clusters of a network
 C) A node that has many outgoing links to authoritative nodes D) A node with no connections
15. What is the primary purpose of a web crawler?
 A) To display search engine results B) To automatically discover and download web pages
 C) To generate web content D) To create website designs
16. What is incremental crawling?
 A) Crawling only newly updated or modified web pages B) Crawling all pages from scratch each time
 C) Ignoring duplicate pages D) Crawling only homepages of websites
17. What is the main goal of sentiment analysis?
 A) To determine the factual accuracy of text B) To classify text as positive, negative, or neutral
 C) To summarize long documents D) To detect spam messages
18. Which of the following is an example of opinion mining?
 A) Identifying the topic of a news article B) Extracting emotions from product reviews
 C) Translating text from one language to another D) Finding plagiarism in research papers
19. What is the primary goal of web usage mining?
 A) Extracting hidden patterns from web log data B) Analyzing website structure
 C) Extracting text content from web pages D) Improving website aesthetics
20. Which of the following is a major source of data for web usage mining?
 A) Web server logs B) HTML code
 C) Multimedia content D) CSS files

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Outline various data formats used for association rule mining.	8	L*2	1	1
	b) Explain an Apriori algorithm to generate frequent item sets.	8	L2	1	1
2.	a) Explain the applications of Apriori algorithm in web mining.	8	L2	1	1
	b) Apply GSP algorithm for frequent itemset generation.	8	L3	2	2
3.	a) Explain any 4 techniques used in data mining.	8	L2	1	1
	b) Explain class association rule mining.	8	L2	2	2
4.	a) What are the models of Information retrieval? Explain each with an example.	8	L2	3	1

Unit – II

b)	What is web crawling? Explain in detail the procedure of Web crawling.	8	L2	4	1
5. a)	Explain the process of meta-search in the Web search process.	8	L2	3	1
b)	Illustrate page ranking algorithm and HITS algorithm.	8	L2	4	1
6. a)	Explain the various steps in text pre-processing.	8	L2	3	1
b)	Differentiate between stemming and lemmatization.	8	L3	4	2

Unit – III

7. a)	What are key elements of Web usage data preprocessing? Explain.	8	L2	5	1
b)	Explain sentiment classification.	8	L2	5	1
8. a)	Explain the steps involved in feature-based sentiment analysis.	8	L2	5	1
b)	Demonstrate cluster analysis and visitor segments in Web-mining.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
